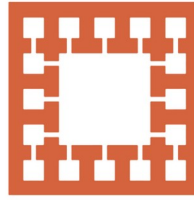


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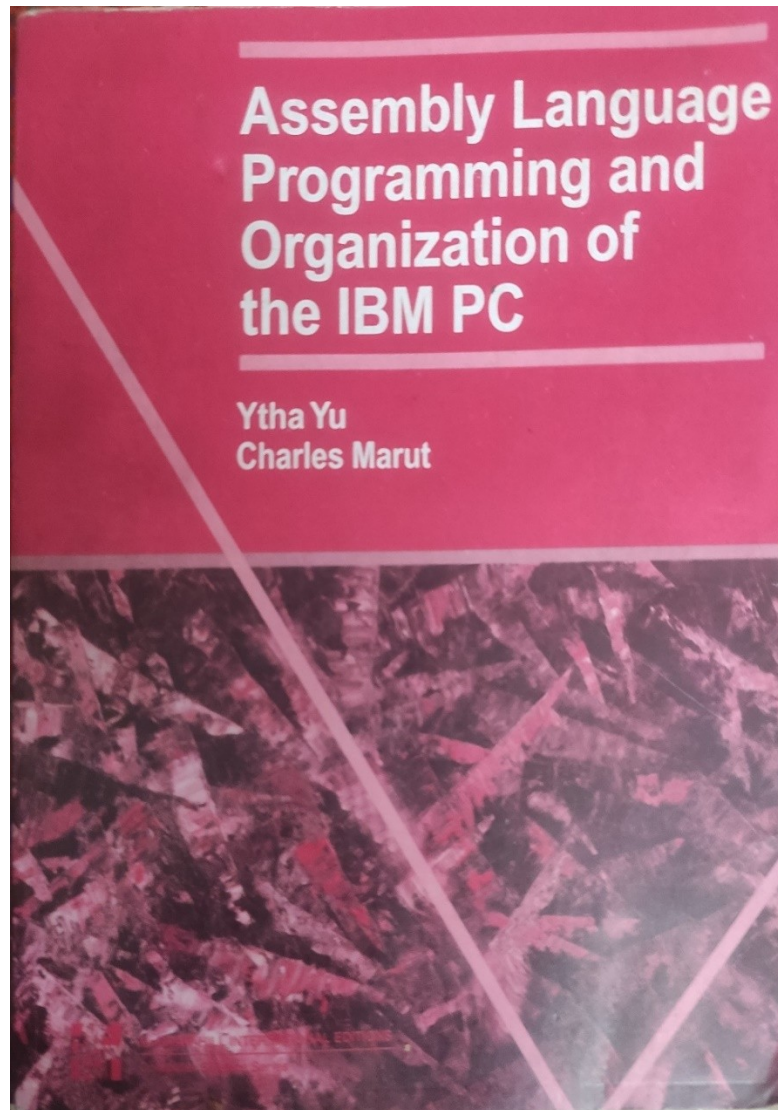
Basic Discussion

Course Title: **Microprocessor and Assembly Language**

Course No: **CSE 3205**

Prepared By:
Md. Adnan Sami
Lecturer, Dept. of CSE, VU

Reference Book



Outline

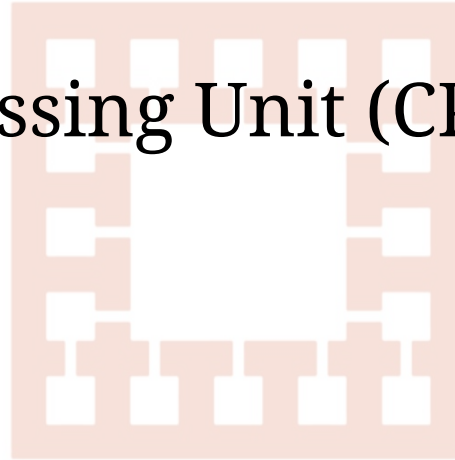
1. Component of a microcomputer system.
2. Why CPU is called the brain of the computer.
3. Operation of Control Unit.
4. Operation of Arithmetic Logic Unit (ALU).



Component of a microcomputer system

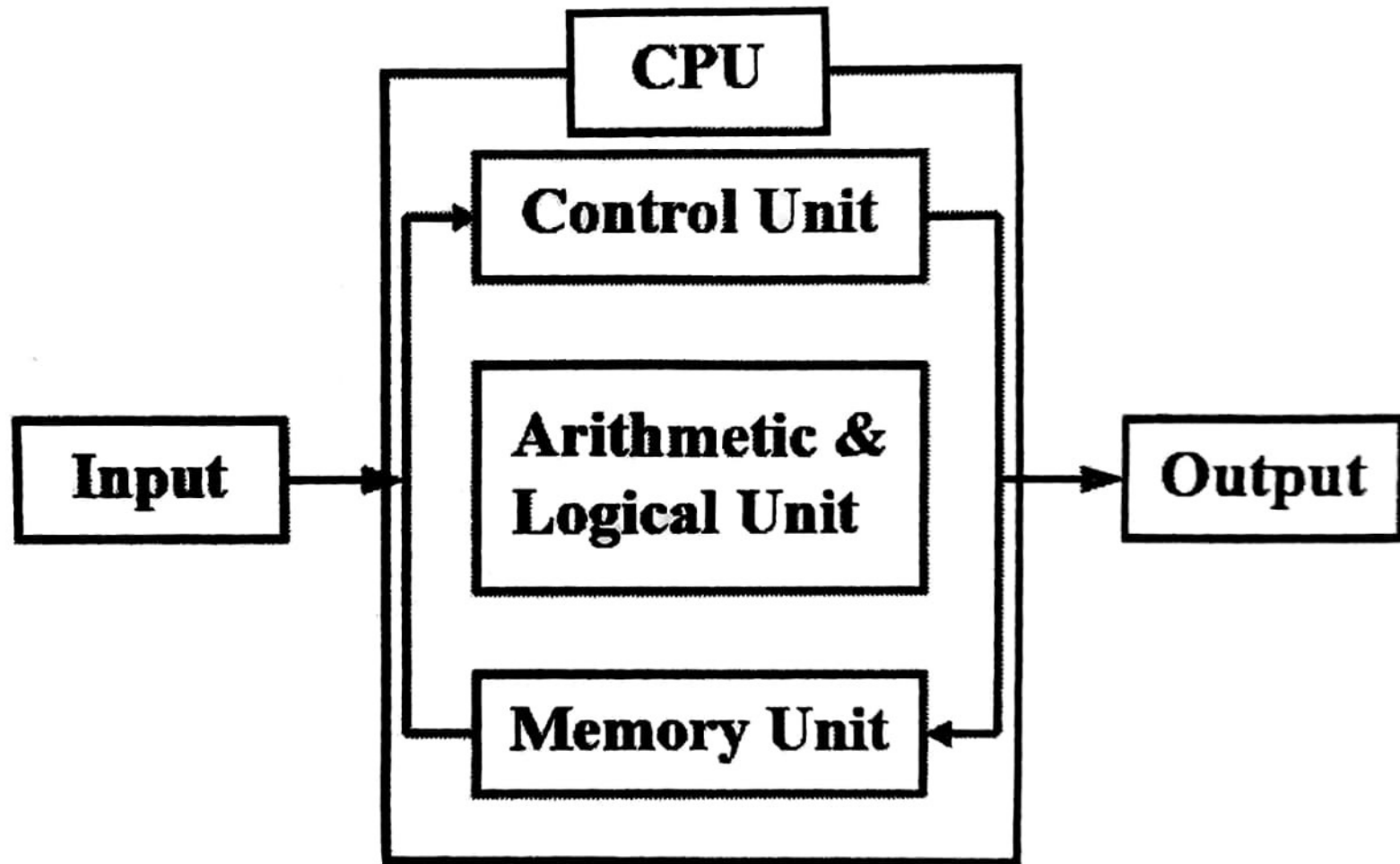
Three parts of computer system.

1. Central Processing Unit (CPU).
2. Memory.
3. I/O Devices.



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Component of a microcomputer system



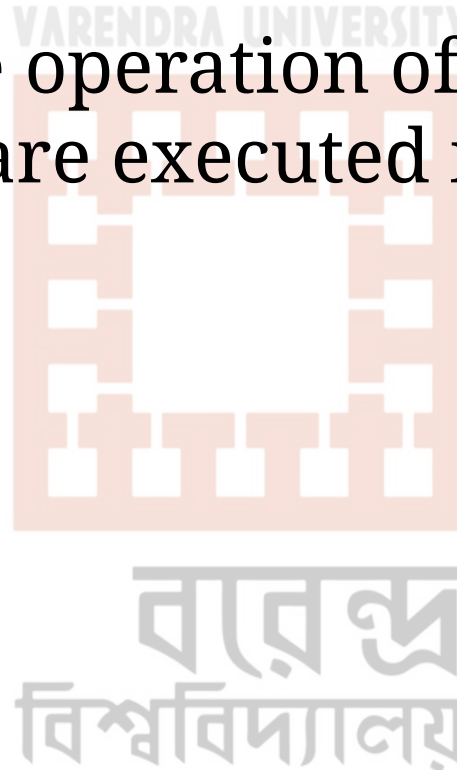
Component of a microcomputer system

Working Principal:

1. **Central processing unit (CPU):** It controls the operation of the computer.
2. **Memory:** It stores data.
3. **I/O Device:** Moves data between the computer and its external environment.

Why CPU is called the brain of the computer

- **CPU** controls the operation of the computer.
- All instructions are executed in CPU.

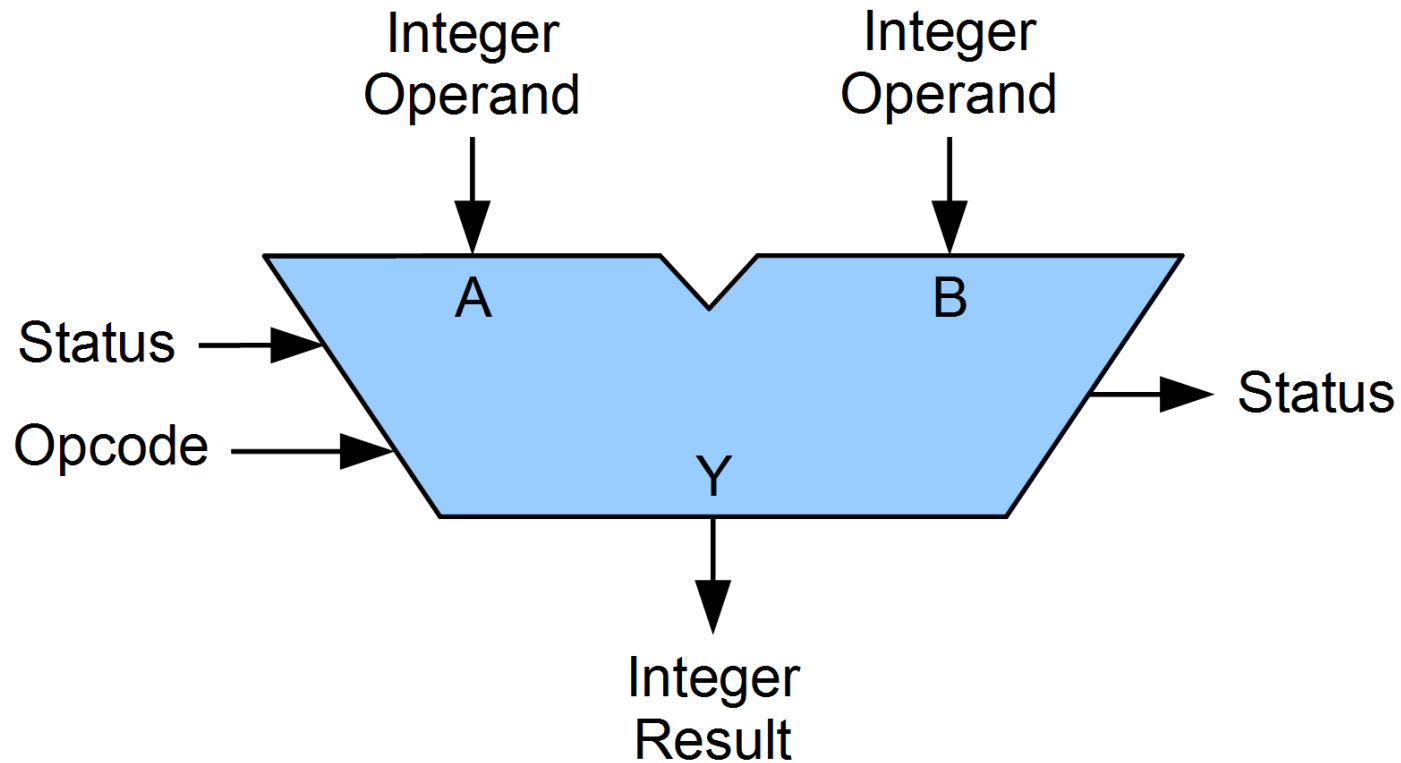


Operation or Function of Control Unit

1. **Fetch:** Control unit fetches instructions from memory that need to be executed.
2. **Decode:** After fetching, control unit decodes the instruction. Decode means understand the binary pattern.
3. **Execution:** After decoding, control unit executes the instruction.

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Operation of Arithmetic Logic Unit (ALU)



Operation of Arithmetic Logic Unit (ALU)

- 1. Arithmetic Operation:** Arithmetic operation like addition, subtraction, multiplication etc. is performed.
- 2. Logical Operation:** Logical operation like AND, OR, NOT is performed.
- 3. Data Manipulation:** Altering or transforming data to make it more manageable and meaningful.